- The C4 model is a formal notation used to document architecture. It was designed to be used at every detail level, including the class level, component/ module level, container/service level, and system level. You can use it to visualize from different perspectives such as deployment.
- To design the architecture of a data product, you need to first capture and understand its architectural drivers.
- Architectural drivers are functional requirements, quality attributes, constraints, and principles.
- Some architectural drivers come from the governance body, some from source or consuming systems, and others from functional requirements and descriptions of the data product.
- Designing the data product should be a collaborative and inclusive exercise.
- To choose from design options, you can use a pro-con-fix or tradeoff analysis exercise.
- Never forget about architectural drivers during the design; you should go back to them regularly.
- To deal with file-based data products, you can take advantage of the platform that should include tools like Airflow.
- Turning a database inside out is an easy way to expose data from systems of record.
- The CQRS pattern is an example of how read models can be created. The most favorable way is by creating a new service with a separate database synchronized using event-based communication.
- You can use the outbox pattern to expose events from your system of record.
- A data mesh implementation does not exclude the possibility of using a data lake; these concepts can be complementary.